# Indoor Scene Classification by Incorporating Predicted Depth Descriptor

Yingbin Zheng[1], Jian Pu[2], Hong Wang[1], and Hao Ye[1]*

[1] Shanghai Advanced Research Institute, Chinese Academy of Sciences
[2] Shanghai Key Lab for Trustworthy Computing, School of Computer Science and Software Engineering, East China Normal University, Shanghai, China
{zhengyb,wang_hong,yeh}@sari.ac.cn, jianpu@sei.ecnu.edu.cn

**Abstract.** Depth cue is crucial for perception of spatial layout and understanding the cluttered indoor scenes. However, there is little study of leveraging depth information within the image scene classification systems, mainly because the lack of depth labeling in existing monocular image datasets. In this paper, we introduce a framework to overcome this limitation by incorporating the predicted depth descriptor of the monocular images for indoor scene classification. The depth prediction model is firstly learned from existing RGB-D dataset using the multiscale convolutional network. Given a monocular RGB image, a representation encoding the predicted depth cue is generated. This predicted depth descriptors can be further fused with features from color channels. Experiments are performed on two indoor scene classification benchmarks and the quantitative comparisons demonstrate the effectiveness of proposed scheme.

**Keywords:** Indoor scene classification, predicted depth descriptor

## 1 Introduction

Scene understanding is a challenge problem in computer vision and multimedia research. As an important step for scene understanding, scene classification considers the semantic concepts and assigns images into their associated scene categories. With the development of internet and storage technologies in last decade, a few scene-centric datasets, such as Fifteen Natural Scene [1], MIT Indoor Scene [2] and SUN Database [3], have been proposed and a significant amount of progress has gone into recognizing scene via various image descriptors such as low-level features, mid-level features, and learning-based representations.

Most of previous scene classification frameworks are based on the color channels of the images. Recently, with the progress of the camera sensor technology, people can get affordable RGB-D cameras such as Microsoft Kinect and Intel RealSense in the consumer market and take images with reliable depth information. These sensors have significantly boosted several research problems' revolution,

* Corresponding author.

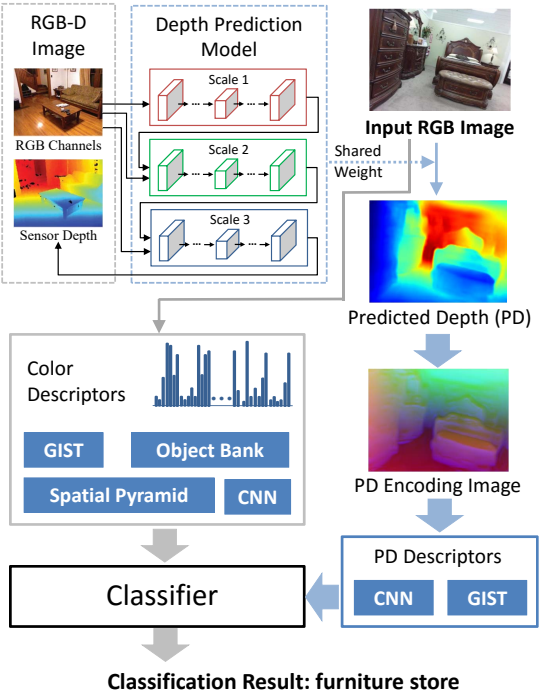



**Fig. 1.** Overview of our frameworks. Depth prediction model is trained and employed to generate the predicted depth for an input RGB images. The features are extracted with predicted depth and color cues, and then applied to scene classifier.

e.g., pose recognition [4] and image segmentation [5]. There are also many works to assign semantic labels to an RGB-D images (e.g., [5–9]) , and depth information are proved to be a very important cue for perception of spatial layout and understanding the cluttered indoor scenes. However, all of these approaches are trained using RGB-D datasets and then also tested in RGB-D images. Since the amount of RGB-D cameras and RGB-D images are order-of-magnitude smaller compared with that of traditional monocular RGB cameras, in real world application most of the images required to get the scene category are color images without depth channel. On the other hand, previous scene classification works seldom consider the depth perception within their feature designs. We believe that incorporating depth cue has the potential to further improve current color based scene classification approaches.

In this paper, we propose a framework by incorporating the predicted depth cue into descriptors of the monocular images for indoor scene classification. An overview of the framework is illustrated in Figure 1. Our work is inspired by the progress on the depth estimation of monocular images. Starting with existing RGB-D image dataset, a prediction model is trained and employed to generate the predicted depth. Then the depth encoding strategies are employed to

map the raw predicted depth information into another intermediate channels. These intermediate channels form the regular RGB images and can be encoded into the image descriptors by employing different types of feature extractors. Finally, a multi-class classifier is applied on predicted depth features as well as the color features. The idea of constructing depth based representations for scene classification has been taken in [8–13]. Our novelty lies in the framework that combines depth prediction and descriptor generation upon the prediction. A framework that does pixel-wise prediction is not new, nor is the descriptor based on intermediate channels. But to the best of our knowledge, such a combination has not been proposed before for scene classification. Compared with these approaches using depth maps captured directly from the RGB-D sensors (namely, sensor depth), the predicted depth descriptors are extracted explicitly from RGB channels and can be applied to regular monocular images. The utilization of predicted depth makes our representation not limited by the RGB-D application and more widely available than the existing works. Although there are some loss on the depth metric, we show that the predicted depth features based on depth prediction model are with similar discriminant ability as the depth features obtained directly from depth sensors. Quantitative study shows predicted depth descriptors are consistently complementary with the color features.

The remainder of this paper is organized as follows. Section 2 reviews the related works on scene categorization and RGB-D applications. Section 3 describes our proposed approach. In Section 4, we demonstrate the quantitative study on scene categorization datasets. And finally, we conclude our work in Section 5.

## 2 Related Works

**Scene classification.** A number of attempts to improve the scene classification task exist in the literature. As stated above, we can roughly summarize the descriptors for scene categorization into three groups: low-level features, mid-level features and learning-based representations. The low-level features try to acquire the global information (e.g., [14]) or appearance of local patches (e.g., [1]). Mid-level features focus on the meaning of image regions, for instance, [15] aim to describe objects in scene images, while [16, 17] try to extract discriminative part representations. Recently, deep learning based approaches are proposed (e.g., [18–20]). Their models are trained from large scale image datasets [19, 21] and achieve state-of-the-art performance. Our approach is different from the above approaches since we get a representation to describe images' depth information explicitly.

**RGB-D.** There are many previous works to assign semantic label to an RGB-D images. For example, [5, 6] perform semantic segmentation on pixel level; [7] performs object detection on region level. Probably most similar to our approach is the works of [8] and [9] on using depth maps for scene categorization. Importantly, our framework is not limited to the RGB-D application but can be applied to regular color images.

# 3 Framework

We now elaborate the construction of our framework. We start by introducing the predicted depth image generation, followed by extracting the representation after the depth processing. Then predicted depth descriptors as well as the features directly extracted from color channels are employed for scenes classification.

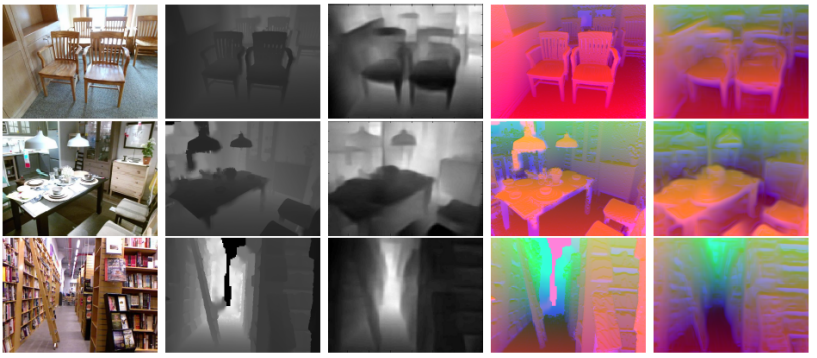
## 3.1 From Input Image to Predicted Depth

Single image depth estimation is a long-standing problem and there exists a large body of literature [22–28]. Recently, several works have used the deep neural networks and led to major advances. As the focus of this paper is to exploit the adaptability of predicted depth cue for scene classification but not to design the depth prediction system, we use the state-of-the-art depth prediction model by multiscale convolutional network [24]. Here we briefly describe the prediction strategy.

First, all the images are resized to the resolution of $320 \times 240$ pixels and set as the input of a three scales network. Scale 1 contains 5 convolutional layers and 2 fully connected layers, which predicts a spatially varying set of features representing full-image view. Scale 2 is to predict at a mid-level resolution, by applying 5 convolutional layers and setting both input image and output features of the first scale as input. Scale 3 of the model employs a 4-convolutional-layer structure to refines the predictions. The multi-scale approach contains both coarse prediction from the entire image and fine prediction locally. The training procedure can be separated into two phases, which are the joint training of Scale 1 and 2, and further training of Scale 3 with frozen of previous scales. Both phases use stochastic gradient descent to learn the parameters. Given an image, the depth prediction model can output a $147 \times 109$ depth map and we bilinearly up-sample it back to the original image resolution.

## 3.2 From Predicted Depth to Descriptor

With the predicted depth, we are now able to build depth models for images with only color channels. A central issue in building a successful model is the design of the representation. There are a few approaches to make use of depth image based on past experiences on scene analysis and intuitions on what may be useful. One encoding method is to treat the predicted depth as a gray-level image by linearly scaling the depth values to the 0 to 255 range and then put the information into all of the RGB color channels to form an RGB image. Another approach is to map the predicted depth values with three properties of geocentric pose, i.e., horizontal disparity, height above ground, and angle of the pixel's local surface normal [7]. After scaling all three channels to 0 to 255 range, the depth information can be transformed into a color image, which has shown the efficiency in object detection task [7]. We refer the intermediate images as RGB and HHA, respectively. A few examples with the (predicted) depth images and their HHA encoding are illustrated in Figure 2. We can observe the HHA



(a) (b) (c) (d) (e)

**Fig. 2.** Examples of RGB images (a), depth images (b), predicted depth images (c), HHA images based on sensor depth (d) and predicted depth (e).

encoding images from sensor depth and predicted depth are visually similar, although there are some loss for depth prediction. The experiments on SUN RGB-D will examine the effectiveness of this encoding for predicted depth.

Next we build the predicted depth descriptor upon the intermediate channels. Two types of feature are chosen for their performance and representative in scene classification, i.e., GIST and Convolution Neural Network (CNN) based features. GIST [14] is a global visual feature, capturing mainly the global texture distribution in depth images. CNN based features are usually learned from a large-scale image dataset, and proved to be more discriminative and accurate than hand-crafted features. Specifically, we use the structure of Alexnet [29] and VGG19 [30]. The Alexnet is with 5 convolutional layers and 3 fully connected (FC) layers, and VGG19 contains 16 convolutional layers and 3 FC layers. We employ the pretrained models learned from ImageNet [21] and then fine-tune the network using training set of the scene datasets. It has been shown that activations from the fully connected layers are with semantically meaningful embedding for recognition [18]. Thus, for each input image representing predicted depth, the 4096-dimensional deep features from the last FC layer are used as CNN representation.

### 3.3 From Descriptor to Scene Prediction

The Multi-class linear SVM classifier is adopted for the depth features. As displayed in Figure 2, images representing the depth cue will lose details of the visual appearance. In order to benefit from the complementarity of regular features in color channels, we use the feature fusion to combine their respective classifier scores. Linear SVM is also applied on the color features. The late fusion strategy is employed as its efficiency in previous scene classification frameworks such as [31, 32]. During the test phase, an input image receives a set of scores $S^d = (s_1^d, ..., s_n^d)$ from depth descriptor's classifier ($n$ is the number of scene categories), and $S^{rgb} = (s_1^{rgb}, ..., s_n^{rgb})$ from color feature's classifier. The confidences

of classifier that the image belongs to category $i$ are computed by the softmax transformation of the scores: $c_i^d = \frac{\exp(s_i^d)}{\sum_{k=1}^n \exp(s_k^d)}$, $c_i^{rgb} = \frac{\exp(s_i^{rgb})}{\sum_{k=1}^n \exp(s_k^{rgb})}$. The overall confidence for category $i$ is done by fusion of these confidences: $c_i = c_i^d \times c_i^{rgb}$. Finally, the image is assigned to the category with the highest confidence after multiplication. Experimental results will show the discriminative ability of the combination in next section.

## 4 Experiments

We evaluate the proposed framework on two indoor scene classification benchmarks: SUN RGB-D scene categorization [9] and the MIT Indoor Scene [2]. SUN RGB-D contains 10,335 images captured by RGB-D sensors (e.g., Kinect and Intel RealSense). Some of images are collected from NYU depth v2 [5], Berkeley B3DO [33], and SUN3D [34]. There are 19 scene categories with more than 80 images and we follow the standard train, validate and test split in the scene categorization benchmark. The MIT Indoor Scene contains 15,620 images labeled over 67 classes. We adopt the official train/test split to use 80 images per class for training and 20 for testing. The performance is evaluated by calculating the average accuracy on each category of both datasets. Throughout the experiments, we use the highly efficient multi-class linear LIBSVM for scene classification.

### 4.1 SUN RGB-D

**Encoding strategy.** We first examine the effectiveness of the HHA encoding for predicted depth. Both HHA and RGB encoding approaches are tested with Alexnet structure. The accuracy of RGB encoding is 19.4%, which is much lower than that of HHA (30.3%). We attribute this to the geocentric pose properties of HHA that avoid negative effects on the inaccuracy predicted on metric depth.
**Predicted depth descriptors.** As illustrated in Table 1, there is a gap between GIST and deep learning based features with more than 10% improvement

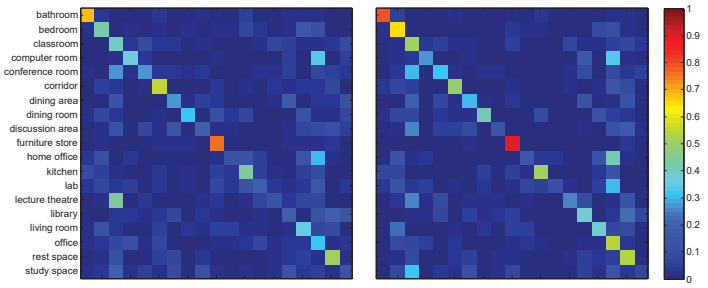
**Table 1.** Comparison for SUN RGB-D using sensor depth images and predicted depth images. Both approaches are with HHA encoding.

(a) Depth descriptor only.

| Depth descriptor | Sensor depth | Predicted depth |
|---|---|---|
| GIST | 20.1% | 18.5% |
| Alexnet | 35.7% | 30.3% |
| VGG19 | 40.7% | 35.2% |

(b) Fusion of depth descriptor with color.

| Depth descriptor | Sensor depth+color | Predicted depth+color |
|---|---|---|
| GIST | 23.0% | 22.8% |
| Alexnet | 41.8% | 39.2% |
| VGG19 | 44.5% | 42.9% |




(a) PD, accuracy=35.2% (b) SD, accuracy=40.7%

**Fig. 3.** Confusion matrices for classification result based on predicted depth (PD) and sensor depth (SD) VGG19 feature in SUN RGB-D.

**Table 2.** Comparisons of state-of-the-arts with only RGB images as inputs.

| Method | This paper | Song et al.[13] | Wang et al.[12] | Zhu et al.[11] | Liao et al.[10] |
|---|---|---|---|---|---|
| Result | **42.9%** | 42.7% | 40.4% | 37.0% | 36.1% |

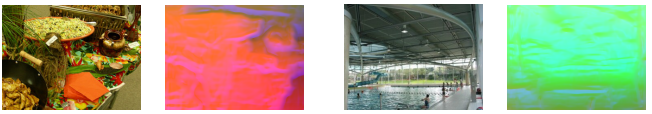
on both the predicted depth and the fusion. The VGG19 network performs better than Alexnet, which indicates that a deeper network for extracting features might improve the discrimination of the depth feature.

**Predicted depth vs sensor depth.** Overall, the predicted depth features achieve very impressive performance comparing with sensor depth. For example, the GIST feature with predicted depth produces an accuracy of 18.5%, with only a slight loss from the same feature representing sensor depth cue (20.1%). The confusion matrices of predicted depth and sensor depth VGG19 features are illustrated in Figure 3. We see that although the average accuracy decreases from 40.7% to 35.2%, the confusion matrices between each two categories are with similar distribution. Another observation is the predicted depth VGG19 feature (35.2%) is on par with the feature using sensor depth (35.7%), which verifies that a stronger representation can fill in the gap between predicted depth and sensor depth.

**Fusion of depth and color.** Since our focus is the depth information and the SUN RGB-D website[3] has provided the CNN features from color channels, we use these off-the-shelf color features on the feature combination (the result of GIST and CNN color features are 19.6% and 38.5%). Fusing features always leads to substantial performance gains, i.e., 18.5% to 22.8% for GIST and 35.2% to 42.9% for VGG19, which clearly show the value of joint modeling the predicted depth and color features for scene classification. The gap between sensor and predicted depth are also narrowed after cue fusion (-1.6% to -0.2% for GIST, and -5.5% to -1.6% for VGG19).

**Comparison with state-of-the-arts.** We compare to related methods on SUN RGB-D in Table 2. With only RGB images as inputs, our predicted depth descriptors outperforms previous approaches, which confirm the effectiveness of

[3] `http://rgbd.cs.princeton.edu/`



(a) Category: buffet (b) Category: poolinside

**Fig. 4.** Challenging examples from two categories in MIT indoor dataset that have low prediction accuracies using our method. (Left: original image; Right: HHA image).

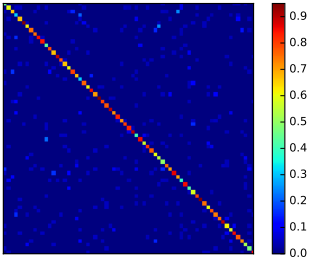



**Fig. 5.** Confusion matrix of MIT indoor using VGG19 feature on both RGB and predicted depth.

using predicted depth as an element for image feature construction. Our results are also better than some approaches considering both RGB and sensor depth channels as inputs, such as [11] (41.5%) and [10] (41.3%). A few works on multi-modal fusion obtain higher accuracy (e.g.[13, 12]) and we believe that predicted depth descriptors is also complementary to the multi-modal fusion since they focus on different levels of information, which will be explored in the future.

### 4.2 MIT Indoor

We also conduct experiments on the MIT Indoor dataset. Note that all the images from MIT Indoor are RGB channels only and without sensor depth information. Applying the predicted depth VGG19 features used in SUN RGB-D obtains the classification accuracy of 26.04%. As the MIT indoor dataset contains richer visual elements and the inter-category variance of depth layout between different classes is very little for the HHA images, only employing the predicted depth information may be not enough. Figure 4 shows two challenge examples. Thus, we fuse the predicted depth features with the visual features directly from color channels. Four popular RGB descriptors are selected, i.e., GIST [14], Spatial Pyramid [1], Object Bank [15], and VGG19 [30], which are corresponding to low-level feature (globally and locally), mid-level feature and learning-based feature. Table 3 gives the fusion results. Incorporating predicted depth feature boosts the performance with consistently improvement, which indicates the complementarity with predicted depth features and color features. Figure 5 illustrates the pre-category results for fusion of predicted depth and RGB VGG19 features with average accuracy of 70.57%.

**Table 3.** Classification accuracy for MIT Indoor dataset with color descriptor only and fusion of color and predicted depth descriptors.

| | Color | Color+predicted depth |
|---|---|---|
| GIST [14] | 28.66% | 33.98% |
| Spatial Pyramid [1] | 34.42% | 42.41% |
| Object Bank [15] | 37.60% | 45.10% |
| VGG19 [30] | 69.33% | 70.57% |

## 5 Conclusion

In this paper, we have introduced a framework to incorporate predicted depth information into image descriptors for scene classification. After setting up the prediction model learned from RGB-D dataset, given an image, an encoding approach is used to transform the predicted depth map to the HHA image. The predicted depth representation is generated by using the GIST or CNN feature on HHA image. The scene classification result is computed by applying linear SVM on predicted depth features and features from color channels. Experimental comparisons with sensor depth features on SUN RGB-D benchmark show the effectiveness of predicted depth representation, and preliminary evaluations on MIT indoor dataset have confirmed the complementarity with the color features.

**Acknowledgments.** This work was supported in part by grants from NSFC (No. 61602459) and STCSM's program (No. 17511101902 and No. 17YF1427100).

## References

1. Lazebnik, S., Schmid, C., Ponce, J.: Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories. In: CVPR. (2006) 2169–2178
2. Quattoni, A., Torralba, A.: Recognizing indoor scenes. In: CVPR. (2009)
3. Xiao, J., Hays, J., Ehinger, K.A., Oliva, A., Torralba, A.: Sun database: Large-scale scene recognition from abbey to zoo. In: CVPR. (2010) 3485–3492
4. Shotton, J., Sharp, T., Kipman, A., Fitzgibbon, A., Finocchio, M., Blake, A., Cook, M., Moore, R.: Real-time human pose recognition in parts from single depth images. Communications of the ACM **56**(1) (2013) 116–124
5. Silberman, N., Hoiem, D., Kohli, P., Fergus, R.: Indoor segmentation and support inference from rgbd images. In: ECCV. (2012)
6. Ren, X., Bo, L., Fox, D.: Rgb-(d) scene labeling: Features and algorithms. In: CVPR. (2012) 2759–2766
7. Gupta, S., Girshick, R., Arbeláez, P., Malik, J.: Learning rich features from rgb-d images for object detection and segmentation. In: ECCV. (2014) 345–360
8. Gupta, S., Arbelaez, P., Malik, J.: Perceptual organization and recognition of indoor scenes from rgb-d images. In: CVPR. (2013) 564–571
9. Song, S., Lichtenberg, S.P., Xiao, J.: Sun rgb-d: A rgb-d scene understanding benchmark suite. In: CVPR. (2015)
10. Liao, Y., Kodagoda, S., Wang, Y., Shi, L., Liu, Y.: Understand scene categories by objects: A semantic regularized scene classifier using convolutional neural networks. In: ICRA. (2016)

11. Zhu, H., Weibel, J.B., Lu, S.: Discriminative multi-modal feature fusion for rgbd indoor scene recognition. In: CVPR. (2016)
12. Wang, A., Cai, J., Lu, J., Cham, T.J.: Modality and component aware feature fusion for rgb-d scene classification. In: CVPR. (2016)
13. Song, X., Herranz, L., Jiang, S.: Depth cnns for rgb-d scene recognition: Learning from scratch better than transferring from rgb-cnns. In: AAAI. (2017)
14. Oliva, A., Torralba, A.: Modeling the shape of the scene: A holistic representation of the spatial envelope. IJCV **42**(3) (2001)
15. Li, L.J., Su, H., Xing, E.P., Fei-Fei, L.: Object bank: A high-level image representation for scene classification semantic feature sparsification. In: NIPS. (2010)
16. Juneja, M., Vedaldi, A., Jawahar, C.V., Zisserman, A.: Blocks that shout: Distinctive parts for scene classification. In: CVPR. (2013) 923–930
17. Doersch, C., Gupta, A., Efros, A.A.: Mid-level visual element discovery as discriminative mode seeking. In: NIPS. (2013) 494–502
18. Razavian, A., Azizpour, H., Sullivan, J., Carlsson, S.: Cnn features off-the-shelf: an astounding baseline for recognition. In: CVPR Workshops. (2014) 806–813
19. Zhou, B., Lapedriza, A., Xiao, J., Torralba, A., Oliva, A.: Learning deep features for scene recognition using places database. In: NIPS. (2014) 487–495
20. Gong, Y., Wang, L., Guo, R., Lazebnik, S.: Multi-scale orderless pooling of deep convolutional activation features. In: ECCV. (2014) 392–407
21. Deng, J., Dong, W., Socher, R., Li, L., Li, K., Fei-Fei, L.: ImageNet: A large-scale hierarchical image database. In: CVPR. (2009)
22. Saxena, A., Sun, M., Ng, A.Y.: Make3d: Learning 3d scene structure from a single still image. IEEE Trans. PAMI **31**(5) (2009) 824–840
23. Liu, B., Gould, S., Koller, D.: Single image depth estimation from predicted semantic labels. In: CVPR. (2010)
24. Eigen, D., Fergus, R.: Predicting depth, surface normals and semantic labels with a common multi-scale convolutional architecture. In: ICCV. (2015) 2650–2658
25. Liu, F., Shen, C., Lin, G.: Deep convolutional neural fields for depth estimation from a single image. In: CVPR. (2015)
26. Zoran, D., Isola, P., Krishnan, D., Freeman, W.T.: Learning ordinal relationships for mid-level vision. In: ICCV. (2015)
27. Wang, P., Shen, X., Lin, Z., Cohen, S., Price, B., Yuille, A.: Towards unified depth and semantic prediction from a single image. In: CVPR. (2015) 2800–2809
28. Chen, W., Fu, Z., Yang, D., Deng, J.: Single-image depth perception in the wild. In: NIPS. (2016)
29. Krizhevsky, A., Sutskever, I., Hinton, G.E.: ImageNet classification with deep convolutional neural networks. In: NIPS. (2012)
30. Simonyan, K., Zisserman, A.: Very deep convolutional networks for large-scale image recognition. In: ICLR. (2015)
31. Pandey, M., Lazebnik, S.: Scene recognition and weakly supervised object localization with deformable part-based models. In: ICCV. (2011)
32. Zheng, Y., Jiang, Y.G., Xue, X.: Learning hybrid part filters for scene recognition. In: ECCV. (2012) 172–185
33. Janoch, A., Karayev, S., Jia, Y., Barron, J.T., Fritz, M., Saenko, K., Darrell, T.: A category-level 3d object dataset: Putting the kinect to work. In: Consumer Depth Cameras for Computer Vision. (2013) 141–165
34. Xiao, J., Owens, A., Torralba, A.: Sun3d: A database of big spaces reconstructed using sfm and object labels. In: ICCV. (2013) 1625–1632